Technical Document #128: Retrieval Augmented Generation - Comprehensive Overvi

Technical Document #128: Retrieval Augmented Generation - Comprehensive Overview

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. It grounds LLM responses in factual, retrieved documents.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Re-ranking models score retrieved documents for relevance to the query. They improve the quality of context provided to the generator.

Semantic search finds documents based on meaning rather than exact keywords. It uses embedding similarity to match queries with relevant content.

Evaluation metrics for RAG include faithfulness, answer relevance, and context recall. They measure the quality of the retrieval-generation pipeline.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Key Takeaways:

- Hallucination detection identifies when LLMs generate factually incorrect information.
- Hybrid search combines keyword-based and semantic search.
- Evaluation metrics for RAG include faithfulness, answer relevance, and context recall.